

Decision Analysis to Inform Screening of the Young (DAISY):
An interactive tool to support evidence-based newborn screening policy decisions

DAISY is a decision analytic model that compares population-based newborn screening with clinical case identification using specified model inputs to project health outcomes. This application is intended to support decision-making regarding screened conditions by groups such as state newborn screening advisory committees.

Instructions for use and a worked example are provided in a peer-reviewed manuscript.

Smith HS, Stein S, Henry MW, Lam WKK, Ream MA, Ojodu J, Jones E, Comeau AM, Tanksley S, Prosser LA, Kemper AR, Grosse SD. Decision Analysis to Inform Screening of the Young: An interactive newborn screening model. *Pediatrics*. In press.

Web-based application

Users may access the interactive web-based application at <https://hadleyssmith.github.io/DAISY/>

Alternatively, users may run the R code to run the app natively on their own device by following the instructions below.

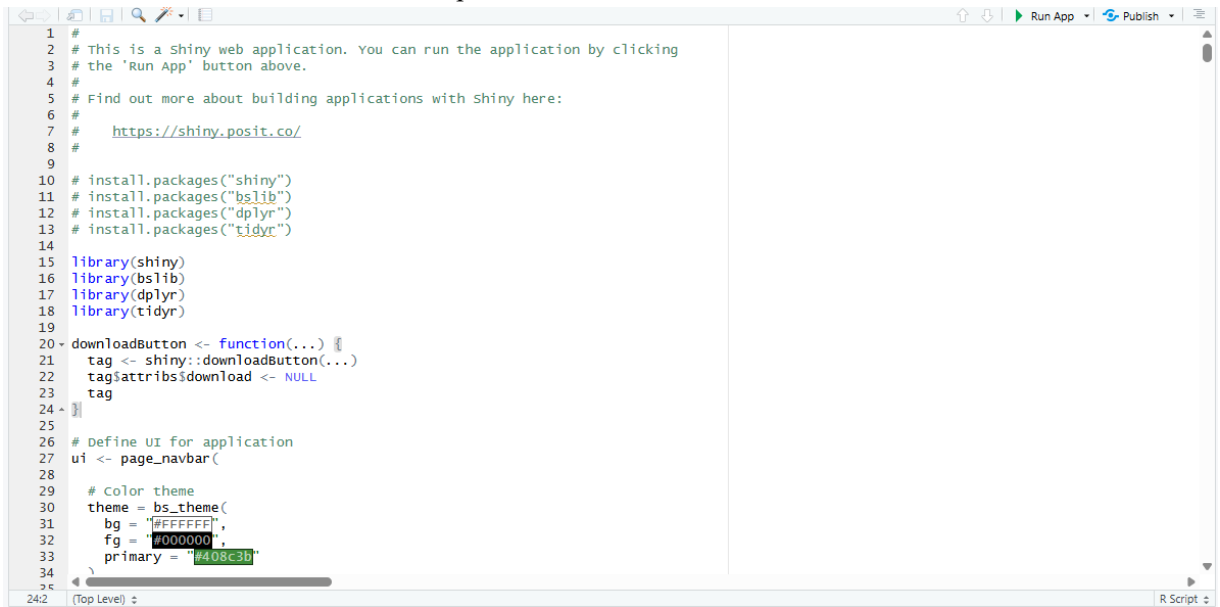
Installation requirements

- Shiny
- Bslib
- Dplyr
- TidyR

Instructions to run natively

- 1.) If R and R studio are not installed onto your machine, please download from the links below
 - a. Link to download R: <https://cran.rstudio.com/>
 - i. Please click on your operating system
 1. For windows: Choose base to be directed to R file download
 2. For macOS: Click the first file available
 - b. Link to download R studio: <https://posit.co/download/rstudio-desktop>
- 2.) Install both R and R studio before moving on to next steps
- 3.) Navigate to DAISY's osf.io page or click on this link: <https://doi.org/10.17605/OSF.IO/R5KE8>
- 4.) Click "Model code" located on DAISY osf.io page, and download the DAISY.R file
- 5.) Use R to open the file

a. Your screen should look similar to the picture below



```
1 #
2 # This is a shiny web application. You can run the application by clicking
3 # the 'Run App' button above.
4 #
5 # Find out more about building applications with shiny here:
6 #
7 # https://shiny.posit.co/
8 #
9
10 # install.packages("shiny")
11 # install.packages("bslib")
12 # install.packages("dplyr")
13 # install.packages("tidyr")
14
15 library(shiny)
16 library(bslib)
17 library(dplyr)
18 library(tidyr)
19
20 downloadButton <- function(...) {
21   tag <- shiny::downloadButton(...)
22   tag$attribs$download <- NULL
23   tag
24 }
25
26 # Define UI for application
27 ui <- page_navbar(
28
29   # color theme
30   theme = bs_theme(
31     bg = "#FFFFFF",
32     fg = "#000000",
33     primary = "#408c3b"
34   )
35 )
```

b. If you are new to R, please remove the “#” on the specified lines below:

- i. “# install.packages("shiny")”
- ii. “# install.packages("bslib")”
- iii. “# install.packages("dplyr")”
- iv. “# install.packages("tidyr")”

1. If you have used R and have these packages installed or have run this application before,
do not remove the “#” in the specified lines

6.) Click the “Run App” button to activate the shiny app